MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization and first-order optimization

Section 3.5: Chain rule as matrix multiplication

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Unit 1 used matrix multiplication to represent composition of linear maps: applying B first and then A gives $A(B\mathbf{x}) = (AB)\mathbf{x}$. For differentiable nonlinear maps, the same idea reappears locally. The local linear maps compose, so their Jacobian matrices multiply.

The chain rule

The chain rule describes derivatives of composite functions.

Compositions

Definition: Composite Function

If $G : \mathbb{R}^n \to \mathbb{R}^p$ and $F : \mathbb{R}^p \to \mathbb{R}^m$ are multivariable functions, we can form a new composite function as follows. Take an input in $\mathbb{R}^n$, and then apply G to the input to obtain a vector in $\mathbb{R}^p$. Apply F to this vector to obtain a vector $\mathbb{R}^m$. This is the output of the composite function. In other words, our input is fed through the following diagram

$$\mathbb{R}^n \xrightarrow{G} \mathbb{R}^p \xrightarrow{F} \mathbb{R}^m.$$

We write the composite function as $F \circ G$ (the symbol $\circ$ should be read as "composed with"). In symbols, the composite function is defined as

$$(F \circ G)(\mathbf{x}) = F(G(\mathbf{x})).$$

Note

Note that the composite $F \circ G$ is *only* defined when the output of G is a possible input to F, i.e., the output of G is $\mathbb{R}^p$, which is the space of inputs to F.

Unit 1 used matrix multiplication to represent composition of linear maps: applying B first and then A gives $A(B\mathbf{x}) = (AB)\mathbf{x}$.

Activity: Composing two matrix maps (U1-LO5, U1-LO6)

Let $T_1 : \mathbb{R}^2 \to \mathbb{R}^3$ and $T_2 : \mathbb{R}^3 \to \mathbb{R}^2$ be linear maps defined by their component functions as follows:

$$T_1(x, y) = \begin{bmatrix} 2x - y \\ 3y \\ x + y \end{bmatrix} \qquad T_2(u, v, w) = \begin{bmatrix} u + v - w \\ 2u - v \end{bmatrix}.$$

Find the matrices A_1 and A_2 associated with these maps, so that $T_1 = T_{A_1}$ and $T_2 = T_{A_2}$ (recall the notation from Matrix map).

We recall the method of Every linear map has a matrix. In particular, we obtain the first column of A_1 from the entries of $T_1(\mathbf{e}_1)$, and the second column from $T_1(\mathbf{e}_2)$. We calculate that

$$T_1(\mathbf{e}_1) = T_1(1,0) = \begin{bmatrix} 2\cdot 1 - 0 \\ 3\cdot 0 \\ 1 + 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

and that

$$T_1(\mathbf{e}_2) = T_1(0,1) = \begin{bmatrix} 2\cdot 0 - 1 \\ 3\cdot 1 \\ 0 + 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}.$$

So the matrix A_1 is equal to

$$A_1 = \begin{bmatrix} 2 & -1 \\ 0 & 3 \\ 1 & 1 \end{bmatrix}.$$

Similarly, we calculate that

$$T_2(\mathbf{e}_1) = T_2(1,0,0) = \begin{bmatrix} 1 + 0 - 0 \\ 2\cdot 1 - 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

that

$$T_2(\mathbf{e}_2) = T_2(0,1,0) = \begin{bmatrix} 0 + 1 - 0 \\ 2\cdot 0 - 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

and that

$$T_2(\mathbf{e}_3) = T_2(0,0,1) = \begin{bmatrix} 0+0-1 \\ 2\cdot 0 - 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

So the matrix A_2 is given by

$$A_2 = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -1 & 0 \end{bmatrix}.$$

Activity: Composing two matrix maps (U1-LO5, U1-LO6)

Let $T_1 : \mathbb{R}^2 \to \mathbb{R}^3$ and $T_2 : \mathbb{R}^3 \to \mathbb{R}^2$ be linear maps defined by their component functions as follows:

$$T_1(x,y) = \begin{bmatrix} 2x-y \\ 3y \\ x+y \end{bmatrix} \qquad T_2(u,v,w) = \begin{bmatrix} u+v-w \\ 2u-v \end{bmatrix}.$$

Calculate the composite map $T_2 \circ T_1$ and express it as a matrix map from $\mathbb{R}^2$ to $\mathbb{R}^2$.

Recalling Composition shape check (U1-LO3, U1-LO6), we have that $T_2 \circ T_1 = T_{A_2 A_1}$. Now

$$
\begin{aligned}
A_2 \cdot A_1 &= \begin{bmatrix} 1 & 1 & -1 \\ 2 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 3 \\ 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 \cdot 2 + 1 \cdot 0 & 1 \cdot (-1) + 1 \cdot 3 \\ \quad + (-1) \cdot 1 & \quad + (-1) \cdot 1 \\ 2 \cdot 2 + (-1) \cdot 0 & 2 \cdot (-1) + (-1) \cdot 3 \\ \quad + 0 \cdot 1 & \quad + 0 \cdot 1 \end{bmatrix} \\
&= \begin{bmatrix} 2 + 0 - 1 & -1 + 3 - 1 \\ 4 + 0 + 0 & -2 - 3 + 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 \\ 4 & -5 \end{bmatrix}
\end{aligned}
$$

Therefore, the composite map $T_2 \circ T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ can be expressed as:

$$
(T_2 \circ T_1) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + y \\ 4x - 5y \end{bmatrix}
$$

Here is a nonlinear example of the composition of two functions.

Activity: Composing nonlinear maps in both orders (U3-LO1)

Consider $F : \mathbb{R}^2 \to \mathbb{R}^2$ and $G : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$F(u, v) = (uv, u + v) \quad \text{and} \quad G(x, y) = (e^{xy}, x - y).$$

Find expressions for $(F \circ G)(x, y)$ and $(G \circ F)(u, v)$.

We calculate that

$$\begin{aligned}(F \circ G)(x, y) &= F(G(x, y)) \\ &= F(e^{xy}, x - y) \\ &= (e^{xy}(x - y), e^{xy} + (x - y)) \\ &= (e^{xy}x - e^{xy}y, e^{xy} + x - y),\end{aligned}$$

and that

$$\begin{aligned}(G \circ F)(u, v) &= G(F(u, v)) \\ &= G(uv, u + v) \\ &= (e^{uv(u+v)}, uv - (u + v)) \\ &= (e^{u^2v+uv^2}, uv - u - v).\end{aligned}$$

Warning

Observe that $F \circ G$ and $G \circ F$ are very different functions. We must therefore be careful that we compose functions in the correct order.

The order matters. The rightmost Jacobian matrix acts first on the input change.

Activity: Composition with a parametrized curve (U3-LO1)

Consider $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ and $\mathbf{p} : \mathbb{R}^3 \to \mathbb{R}^2$ defined by

$$\mathbf{r}(t) = (t, \cos(t), \sin(t)) \quad \text{and} \quad \mathbf{p}(x, y, z) = (y, z)$$

Find an expression for $(\mathbf{p} \circ \mathbf{r})(t)$.

We calculate that

$$\begin{aligned}(\mathbf{p} \circ \mathbf{r})(t) &= \mathbf{p}(\mathbf{r}(t)) \\ &= \mathbf{p}(t, \cos(t), \sin(t)) \\ &= (\cos(t), \sin(t)).\end{aligned}$$

Activity: Composition with a parametrized curve (U3-LO1)

Consider $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ and $\mathbf{p} : \mathbb{R}^3 \to \mathbb{R}^2$ defined by

$$\mathbf{r}(t) = (t, \cos(t), \sin(t)) \quad \text{and} \quad \mathbf{p}(x, y, z) = (y, z)$$

Is the composition $\mathbf{r} \circ \mathbf{p}$ defined? Why or why not?

For the composition $\mathbf{r} \circ \mathbf{p}$ to be defined, the outputs of $\mathbf{p}$ must be contained in the inputs of $\mathbf{r}$. But $\mathbf{p}$ outputs points in $\mathbb{R}^2$, whereas $\mathbf{r}$ takes in points in $\mathbb{R}$ as input, so the composition does not make sense.

Chain rule for one-parameter inputs

Recall the chain rule from single-variable calculus, that for two functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$, we have

$$(f \circ g)'(t) = (f' \circ g)(t)g'(t).$$

The next theorem is the component form of the chain rule.
Recall that for a function $f(x, y)$, $D_1 f = f_x = \frac{\partial f}{\partial x}$, and that $D_2 f = f_y = \frac{\partial f}{\partial y}$.

Theorem: Chain Rule for One Independent Variable

Consider a scalar-valued function $f : \mathbb{R}^2 \to \mathbb{R}$, and a vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ of a single variable. Then the composition $f \circ \mathbf{r} : \mathbb{R} \to \mathbb{R}$ is a scalar-valued function of a single variable. If $\mathbf{r}$ is differentiable at t_0, and f is differentiable at $\mathbf{r}(t_0)$, then $f \circ \mathbf{r}$ is differentiable at t_0 and

$$(f \circ \mathbf{r})'(t_0) = (D_1 f \circ \mathbf{r})(t_0)r_1'(t_0) + (D_2 f \circ \mathbf{r})(t_0)r_2'(t_0),$$

where r_1 and r_2 are the components of $\mathbf{r}$.

Why is this true?

We note that the right hand side of the equation can be written as $L \circ \mathbf{r} - f(\mathbf{r}(t_0))$, where L is the linear approximation of f at $\mathbf{r}(t_0)$. Because f is differentiable, we know that

$$\lim_{\mathbf{x}\to\mathbf{r}(t_0)} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{r}(t_0)\|} = 0.$$

Now

$$\lim_{t\to t_0} \frac{\|\mathbf{r}(t) - \mathbf{r}(t_0)\|}{t - t_0} = \left\|\lim_{t\to t_0} \frac{\mathbf{r}(t) - \mathbf{r}(t_0)}{t - t_0}\right\| = \|\mathbf{r}'(t_0)\|.$$

Using these two equations (substituting $\mathbf{r}(t)$ for $\mathbf{x}$ in the first equation), we find that

$$\begin{aligned}\lim_{t\to t_0} \frac{f(\mathbf{r}(t)) - L(\mathbf{r}(t))}{t - t_0} &= \lim_{t\to t_0} \frac{f(\mathbf{r}(t)) - L(\mathbf{r}(t))}{\|\mathbf{r}(t) - \mathbf{r}(t_0)\|} \frac{\|\mathbf{r}(t) - \mathbf{r}(t_0)\|}{t - t_0} \\ &= 0 \cdot \|\mathbf{r}'(t_0)\| = 0.\end{aligned}$$

But recalling that

$$L(x, y) = f(\mathbf{r}(t_0)) + D_1 f(\mathbf{r}(t_0))(x - r_1(t_0)) + D_2 f(\mathbf{r}(t_0))(y - r_2(t_0)),$$

we conclude that

$$\begin{aligned}\lim_{t\to t_0} \frac{f(\mathbf{r}(t)) - f(\mathbf{r}(t_0))}{t - t_0} &= \lim_{t\to t_0} \frac{f(\mathbf{r}(t)) - f(\mathbf{r}(t_0))}{t - t_0} - \frac{f(\mathbf{r}(t)) - L(\mathbf{r}(t))}{t - t_0} \\ &= \lim_{t\to t_0} \frac{L(\mathbf{r}(t)) - f(\mathbf{r}(t_0))}{t - t_0} \\ &= \lim_{t\to t_0} D_1 f(\mathbf{r}(t_0)) \frac{r_1(t) - r_1(t_0)}{t - t_0} \\ &\qquad + D_2 f(\mathbf{r}(t_0)) \frac{r_2(t) - r_2(t_0)}{t - t_0} \\ &= f_x(\mathbf{r}(t_0)) r_1'(t_0) + D_2 f(\mathbf{r}(t_0)) r_2'(t_0).\end{aligned}$$

Note

There are several ways to aid in remembering the formula for the chain rule. The first is to simplify the equation, writing it as

$$(f \circ \mathbf{r})'(t) = (\nabla f \circ \mathbf{r})(t) \cdot \mathbf{r}'(t),$$

so that the equation looks more like the chain rule from single variable calculus. The second is the following mnemonic: if we write z for $f \circ \mathbf{r}$ and for f, and write the components of $\mathbf{r}$ as x and y, then the chain rule reads that

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt},$$

so that the symbols dx and dy in the numerators on the right hand side, and the symbols ∂x and ∂y that occur in the denominators appear to 'cancel' like when multiplying fractions.

Activity: Differentiating through two intermediate variables (U3-LO6)

Suppose $z = x^2y - y^2x$, where $x = \sin t$ and $y = e^t$. Compute dz/dt.

First, we need to calculate

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(x^2y - y^2x) = 2xy - y^2$$
$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(x^2y - y^2x) = x^2 - 2yx$$
$$\frac{dx}{dt} = \frac{d}{dt}(\sin t) = \cos t$$
$$\frac{dy}{dt} = \frac{d}{dt}(e^t) = e^t$$

Now we substitute these into the chain rule

formula:

$$\begin{aligned}\frac{dz}{dt} &= (2xy - y^2) \cdot \cos t + (x^2 - 2yx) \cdot e^t \\ &= (2 \sin t \cdot e^t - e^{2t}) \cdot \cos t \\ &\quad + (\sin^2 t - 2e^t \cdot \sin t) \cdot e^t \\ &= 2 \sin t \cdot e^t \cdot \cos t - e^{2t} \cdot \cos t \\ &\quad + e^t \sin^2 t - 2e^{2t} \sin t\end{aligned}$$

We can simplify this further if we want:

$$\begin{aligned}\frac{dz}{dt} &= 2e^t \sin t \cos t - e^{2t} \cos t + e^t \sin^2 t - 2e^{2t} \sin t \\ &= e^t(\sin^2 t + 2 \sin t \cos t) - e^{2t}(\cos t + 2 \sin t) \\ &= e^t(\sin^2 t + \sin 2t) - e^{2t}(\cos t + 2 \sin t)\end{aligned}$$

Therefore,

$$\frac{dz}{dt} = e^t(\sin^2 t + \sin 2t) - e^{2t}(\cos t + 2 \sin t)$$

Activity: Differentiating a function along a curve (U3-LO6)

If $f(x, y) = x^2 + y^2$, and $\mathbf{r}(t) = (\sin t, \cos(2t))$, find $(f \circ \mathbf{r})'(t)$.

We calculate that

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

and that

$$\mathbf{r}'(t) = \begin{bmatrix} \cos t \\ -2\sin(2t) \end{bmatrix}.$$

Now

$$\nabla f \circ \mathbf{r}(t) = \nabla f(\sin t, \cos(2t)) = \begin{bmatrix} 2\sin t \\ 2\cos(2t) \end{bmatrix}$$

Thus

$$\begin{aligned} (f \circ \mathbf{r})'(t) &= \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \\ &= \begin{bmatrix} 2\sin t \\ 2\cos(2t) \end{bmatrix} \cdot \begin{bmatrix} \cos t \\ -2\sin(2t) \end{bmatrix} \\ &= 2\sin t \cos t - 4\sin(2t)\cos(2t). \end{aligned}$$

Chain rule for two independent variables

Theorem: Chain Rule for Two Independent Variables

Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ is a scalar-valued function, and $G : \mathbb{R}^2 \to \mathbb{R}^2$ is a vector-valued function. Then $f \circ G : \mathbb{R}^2 \to \mathbb{R}$ is a scalar-valued function, and the two equations

$$D_1(f \circ G) = (D_1 f \circ G)D_1 G_1 + (D_2 f \circ G)D_1 G_2$$

and

$$D_2(f \circ G) = (D_1 f \circ G)D_2 G_1 + (D_2 f \circ G)D_2 G_2$$

hold.

Why is this true?

Fix the second variable, and apply Chain Rule for One Independent Variable to the first to obtain the formula for $(f \circ G)_x$. Then fix the first variable and apply Chain Rule for One Independent Variable to the second variable to obtain the formula for $(f \circ G)_y$.

Remark

Recall from Partial derivatives, gradients, Hessians, and Jacobian matrices that the Jacobian matrix collects the partial derivatives of a vector-valued map. For G, this gives

$$J_G = \begin{bmatrix} D_1 G_1 & D_2 G_1 \\ D_1 G_2 & D_2 G_2 \end{bmatrix}.$$

Then we can write the pair of equations in Chain Rule for Two Independent Variables as the single vector equation

$$\nabla(f \circ G) = J_G^T(\nabla f \circ G),$$

where we interpret ∇f as a column vector.

The second method is mnemonic. If we let x and y be the variables of f, and u and v be the variables of G, and we write z for $f \circ G$ and for f, and write the components of $G(u, v)$ as x and y, then the chain rule reads that

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}$$

and that

$$\frac{\partial f}{\partial v} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v},$$

so that the partials in the numerator and denominator again appear to 'cancel'.

Activity: Partial derivatives through a change of variables (U3-LO6)

Suppose that

$$z = x^2 + xy - y^2$$

that $x = e^{2s+t}$, and that $y = \ln(t/s)$. Compute

$$\frac{\partial z}{\partial s} \quad \text{and} \quad \frac{\partial z}{\partial t}.$$

First compute the partial derivatives of z in x and y, and the partial derivatives of x and y in t and s.

We calculate that

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(x^2 + xy - y^2) = 2x + y,$$

that

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(x^2 + xy - y^2) = x - 2y,$$

that

$$\frac{\partial x}{\partial s} = \frac{\partial}{\partial s}(e^{2s+t}) = 2e^{2s+t},$$

that

$$\frac{\partial x}{\partial t} = \frac{\partial}{\partial t}(e^{2s+t}) = e^{2s+t},$$

that

$$\frac{\partial y}{\partial s} = \frac{\partial}{\partial s}\left(\ln\left(\frac{t}{s}\right)\right) = \frac{\partial}{\partial s}(\ln(t) - \ln(s)) = -\frac{1}{s},$$

and that

$$\frac{\partial y}{\partial t} = \frac{\partial}{\partial t}\left(\ln\left(\frac{t}{s}\right)\right) = \frac{1}{t}$$

Activity: Partial derivatives through a change of variables (U3-LO6)

Suppose that

$$z = x^2 + xy - y^2$$

that $x = e^{2s+t}$, and that $y = \ln(t/s)$. Compute

$$\frac{\partial z}{\partial s} \quad \text{and} \quad \frac{\partial z}{\partial t}.$$

Now use the chain rule to compute the partial derivatives of z in s and t.

Now we apply the chain rule, from which we conclude that

$$\begin{aligned}\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \\ &= (2x + y) \cdot 2e^{2s+t} + (x - 2y) \cdot (-1/s) \\ &= 4xe^{2s+t} + 2ye^{2s+t} - (x/s) + (2y/s)\end{aligned}$$

Substituting $x = e^{2s+t}$ and $y = \ln(t/s)$:

$$\begin{aligned}\frac{\partial z}{\partial s} &= 4e^{2s+t} \cdot e^{2s+t} + 2\ln(t/s) \cdot e^{2s+t} \\ &\quad - \frac{e^{2s+t}}{s} + \frac{2\ln(t/s)}{s} \\ &= 4e^{4s+2t} + 2e^{2s+t}\ln(t/s) - \frac{e^{2s+t}}{s} + \frac{2\ln(t/s)}{s}\end{aligned}$$

Similarly for $\frac{\partial z}{\partial t}$:

$$\begin{aligned}\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} \\ &= (2x + y) \cdot e^{2s+t} + (x - 2y) \cdot \frac{1}{t} \\ &= 2xe^{2s+t} + ye^{2s+t} + \frac{x}{t} - \frac{2y}{t}\end{aligned}$$

Substituting $x = e^{2s+t}$ and $y = \ln(t/s)$:

$$\begin{aligned}\frac{\partial z}{\partial t} &= 2e^{2s+t} \cdot e^{2s+t} + \ln(t/s) \cdot e^{2s+t} \\ &\quad + \frac{e^{2s+t}}{t} - \frac{2\ln(t/s)}{t} \\ &= 2e^{4s+2t} + e^{2s+t}\ln(t/s) + \frac{e^{2s+t}}{t} - \frac{2\ln(t/s)}{t}\end{aligned}$$

Activity: The chain rule as a Jacobian product (U3-LO6)

Use the Jacobian matrix version of the chain rule to calculate the partial derivatives of the function $f \circ G$, where $f(x, y) = 3x^2 - 2xy + y^2$, and $G(u, v) = (3u + 2v, 4u - v)$.

We compute that

$$\nabla f(x, y) = \begin{bmatrix} 6x - 2y \\ -2x + 2y \end{bmatrix}$$

and that

$$J_G = \begin{bmatrix} D_1\{3u + 2v\} & D_2\{3u + 2v\} \\ D_1\{4u - v\} & D_2\{4u - v\} \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & -1 \end{bmatrix}$$

Since

$$\begin{aligned} \nabla f(3u + 2v, 4u - v) &= \begin{bmatrix} 6(3u + 2v) - 2(4u - v) \\ -2(3u + 2v) + 2(4u - v) \end{bmatrix} \\ &= \begin{bmatrix} 10u + 14v \\ 2u - 6v \end{bmatrix}, \end{aligned}$$

we find that

$$\begin{aligned}
\nabla(f \circ G) &= \begin{bmatrix} 3 & 2 \\ 4 & -1 \end{bmatrix}^T \begin{bmatrix} 10u + 14v \\ 2u - 6v \end{bmatrix} \\
&= \begin{bmatrix} 3 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 10u + 14v \\ 2u - 6v \end{bmatrix} \\
&= \begin{bmatrix} (3)(10u + 14v) + (4)(2u - 6v) \\ (2)(10u + 14v) + (-1)(2u - 6v) \end{bmatrix} \\
&= \begin{bmatrix} 38u + 18v \\ 18u + 34v \end{bmatrix}.
\end{aligned}$$

Activity: Checking the chain rule in SymPy (U3-LO6)

Revisit $f(x, y) = 3x^2 - 2xy + y^2$ and $G(u, v) = (3u + 2v, 4u - v)$ from the preceding calculation. The code computes $J_{f \circ G}$ in two ways: multiplying the Jacobian matrices and differentiating the composite expression directly. As in Checking the Jacobian of a gradient in SymPy (U3-LO3), we put a scalar expression in a one-entry matrix to use `.jacobian()`. The command `sp.simplify` simplifies symbolic expressions, including the entries of a matrix.

```
import sympy as sp

x, y, u, v = sp.symbols("x y u v")
f = 3*x**2 - 2*x*y + y**2
G = sp.Matrix([3*u + 2*v, 4*u - v])

Jf = sp.Matrix([f]).jacobian([x, y])
```

Activity: Checking the chain rule in SymPy (U3-LO6) (continued)

```
JG = G.jacobian([u, v])
Jf_at_G = Jf.subs({x: G[0], y: G[1]})
chain = sp.simplify(Jf_at_G * JG)

f_of_G = f.subs({x: G[0], y: G[1]})
direct = sp.Matrix([f_of_G]).jacobian([u, v])

print(Jf_at_G.shape, JG.shape, chain.shape)
print(chain)
print(sp.simplify(direct - chain))
print(chain.subs({u: 1, v: 0}))
```

Output:

```
(1, 2) (2, 2) (1, 2)
Matrix([[38*u + 18*v, 18*u + 34*v]])
Matrix([[0, 0]])
Matrix([[38, 18]])
```

1. Explain the three shapes in the first output line. Using the composition rule from Matrix multiplication as composition, explain why `JG` is the right-hand factor. Would the reversed product `JG * Jf_at_G` be defined?
2. Why do we substitute the components of `G` into `Jf`? At $(u, v) = (1, 0)$, where must the outer Jacobian be evaluated? Verify the last output line by multiplying the two evaluated matrices.
3. What does the zero matrix in the third output line establish for these functions? How does the row in the second output line relate to the column gradient $\nabla(f \circ G)$ found by hand?

1. The outer function has one output and two inputs, so $J_f(G(u,v))$ has shape 1×2. The inner map has two outputs and two inputs, so J_G has shape 2×2. Their product has shape 1×2, as required for the Jacobian of a scalar-valued function of u, v. The matrix J_G acts first on an input change; $J_f(G(u,v))$ then acts on the intermediate change. The reversed product is undefined because its inner dimensions would be 2 and 1.

2. The input to the outer function is $G(u,v)$, so its derivative must be evaluated there. At $(u,v) = (1,0)$, we have $G(1,0) = (3,4)$. Therefore

$$\begin{aligned} J_{f\circ G}(1,0) &= J_f(3,4)J_G(1,0) \\ &= \begin{bmatrix} 10 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 4 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 38 & 18 \end{bmatrix}. \end{aligned}$$

   Evaluating J_f at $(1,0)$ instead would use the

wrong input to the outer function.

3. The symbolic difference simplifies to the zero row for all u, v, verifying that the direct derivative and the chain-rule product agree for these polynomial functions. The row is the transpose of the gradient:

$$\nabla(f \circ G)(u, v) = \begin{bmatrix} 38u + 18v \\ 18u + 34v \end{bmatrix}.$$

Transposing $J_{f \circ G} = J_f(G)J_G$ gives the equivalent column formula $\nabla(f \circ G) = J_G^T(\nabla f \circ G)$.

Generalized chain rule

Before multiplying Jacobian matrices, check the input and output dimensions of each map.

Theorem: Generalized Chain Rule

Suppose $F : \mathbb{R}^p \to \mathbb{R}^m$ and $G : \mathbb{R}^n \to \mathbb{R}^p$ are differentiable functions. Then $F \circ G : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable, and

$$J_{F \circ G}(\mathbf{x}) = J_F(G(\mathbf{x}))J_G(\mathbf{x}),$$

where the right hand side is defined by matrix multiplications. Equivalently, for $1 \leq i \leq m$,

$$D_j\{F_i \circ G\}(\mathbf{x}) = \sum_{k=1}^{p} (D_k F_i)(G(\mathbf{x}))(D_j G_k)(\mathbf{x}).$$

The order matters. The rightmost Jacobian matrix acts first on the input change.

Note

If f is a scalar-valued function, then the generalized chain rule tells us that

$$\nabla(f \circ G) = J_G^T(\nabla f \circ G)$$

as in the two variable case.

Activity: The chain rule through a three-component map (U3-LO6)

Suppose $f(x, y, z) = 3x^2 - 2xy + 4z^2$, and that $G(u, v) = (e^u \sin v, e^u \cos v, e^u)$. Compute the partial derivatives of $f \circ G$.

We calculate that

$$\nabla f(x, y, z) = \begin{bmatrix} 6x - 2y \\ -2x \\ 8z \end{bmatrix}$$

so that

$$\nabla f(e^u \sin v, e^u \cos v, e^u) = e^u \begin{bmatrix} 6\sin v - 2\cos v \\ -2\sin v \\ 8 \end{bmatrix}.$$

Now

$$J_G(u,v) = \begin{bmatrix} e^u \sin v & e^u \cos v \\ e^u \cos v & -e^u \sin v \\ e^u & 0 \end{bmatrix}$$
$$= e^u \begin{bmatrix} \sin v & \cos v \\ \cos v & -\sin v \\ 1 & 0 \end{bmatrix}.$$

Thus

$$
\begin{aligned}
&\nabla(f \circ G)(u,v) \\
&= J_G(u,v)^T (\nabla f \circ G)(u,v) \\
&= (e^u)^2 \begin{bmatrix} \sin v & \cos v & 1 \\ \cos v & -\sin v & 0 \end{bmatrix} \begin{bmatrix} 6\sin v - 2\cos v \\ -2\sin v \\ 8 \end{bmatrix} \\
&= e^{2u} \begin{bmatrix} 6(\sin v)^2 - 2\sin v \cos v \\ -2\sin v \cos v + 8 \\ 6\sin v \cos v - 2(\cos v)^2 \\ +2(\sin v)^2 \end{bmatrix} \\
&= e^{2u} \begin{bmatrix} 6(\sin v)^2 - 4\sin v \cos v + 8 \\ 4(\sin v)^2 + 6\sin v \cos v - 2 \end{bmatrix}.
\end{aligned}
$$

From this calculation, we can read that the partial derivative of $f \circ G$ in the first variable is equal to

$$e^{2u}(6(\sin v)^2 - 4\sin v \cos v + 8)$$

and the derivative in the second variable is

$$e^{2u}(4(\sin v)^2 + 6\sin v \cos v - 2).$$

Activity: Differentiating a loss through its predictions (U3-LO6)

The squared-error loss in From predictions to a scalar error function (U3-LO1, U2-LO1) is a composition:

$$\mathbf{w} \longmapsto R(\mathbf{w}) = X\mathbf{w} - \mathbf{y} \longmapsto q(R(\mathbf{w})), \qquad q(\mathbf{r}) = \mathbf{r}^T\mathbf{r}.$$

Thus $\mathscr{L} = q \circ R$. The fixed data are still $X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ and $\mathbf{y} = (1, 2, 2)$.

1. Find $J_R(\mathbf{w})$ and $J_q(\mathbf{r})$, including their shapes. Which map is affine, and which is nonlinear?
2. Use the matrix chain rule to obtain $J_{\mathscr{L}}(\mathbf{w})$, and transpose it to obtain $\nabla\mathscr{L}(\mathbf{w})$. Evaluate at $\mathbf{w}_0 = (1, 1)$ and compare with How model weights change the loss (U3-LO3).
3. Let $\mathbf{w}(t) = (1, 1) + t(1, 1)$. Find $\frac{d}{dt}\mathscr{L}(\mathbf{w}(t))$ using the chain rule. Check by first substituting into the loss and then differentiating the resulting one-variable polynomial.

1. The residual map $R : \mathbb{R}^2 \to \mathbb{R}^3$ is affine, with

$$J_R = X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Its shape is 3×2. The squared-length function $q : \mathbb{R}^3 \to \mathbb{R}$ is nonlinear, with

$J_q(\mathbf{r}) = 2\mathbf{r}^T$, a 1×3 row.

2. Evaluate the outer derivative at the residual, then multiply:

$$J_{\mathscr{L}}(\mathbf{w}) = 2(X\mathbf{w} - \mathbf{y})^T X,$$
$$\nabla\mathscr{L}(\mathbf{w}) = 2X^T(X\mathbf{w} - \mathbf{y}).$$

   The row product has shape $(1 \times 3)(3 \times 2) = 1 \times 2$; the gradient is a 2×1 column. At $\mathbf{w}_0$, the residual is $(0, -1, 0)$, so

$$\nabla\mathscr{L}(\mathbf{w}_0) = 2\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}.$$

   This agrees with differentiating the expanded loss. Each gradient component combines the errors from the observations using the corresponding feature column.

3. The chain rule along the parameter line gives

$$
\begin{aligned}
&\frac{d}{dt}\mathscr{L}(\mathbf{w}(t)) \\
&= \nabla\mathscr{L}(1+t, 1+t)^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 6t & -2+6t \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
&= -2 + 12t.
\end{aligned}
$$

Direct substitution gives $\mathscr{L}(1+t, 1+t) = 1 - 2t + 6t^2$, whose derivative is the same. At $t = 0$, the rate is -2; multiplying by a small parameter change $t = 1/10$ recovers the predicted loss change $-1/5$ from Predicting a change in fitting error (U3-LO4).

Note

The generalized chain rule says that local linear maps compose. The matrix formulation records that composition by multiplying Jacobian matrices in the correct order.

A tiny sigmoid block

A small neural-network block can be read as a composition of three steps: an affine map, a coordinatewise nonlinear rule, and another affine map.

Note: Sigmoid and its derivative

Let

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$

The derivative is

$$\sigma'(t) = \sigma(t)(1 - \sigma(t)).$$

When σ is applied to a vector, it is applied coordinate by coordinate.

Activity: A tiny sigmoid block (U3-LO5, U3-LO6, U3-LO7)

Let

$$W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad W_2 = \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \end{bmatrix},$$

and let $\mathbf{b}_1 = \mathbf{0}$ and $\mathbf{b}_2 = \mathbf{0}$.
Define

$$\mathbf{s} = W_1\mathbf{x} + \mathbf{b}_1, \qquad \mathbf{u} = \sigma(\mathbf{s}), \qquad N(\mathbf{x}) = W_2\mathbf{u} + \mathbf{b}_2,$$

where σ is applied coordinatewise.
At the base point

$$\mathbf{a} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

do the following.

1. Compute the pre-activation vector $\mathbf{s}_\mathbf{a} = W_1\mathbf{a} + \mathbf{b}_1$.

Activity: A tiny sigmoid block (U3-LO5, U3-LO6, U3-LO7) (continued)

2. Compute the hidden vector $\mathbf{u_a} = \sigma(\mathbf{s_a})$.
3. Compute $N(\mathbf{a})$.
4. Use $\sigma'(0) = 1/4$ to compute

$$D_{\mathbf{a}} = \operatorname{diag}\big(\sigma'((\mathbf{s_a})_1), \sigma'((\mathbf{s_a})_2), \sigma'((\mathbf{s_a})_3)\big).$$

5. Use the chain rule to compute

$$J_N(\mathbf{a}) = W_2 D_{\mathbf{a}} W_1.$$

6. Use

$$N(\mathbf{a} + \mathbf{h}) \approx N(\mathbf{a}) + J_N(\mathbf{a})\mathbf{h}$$

to predict the output change caused by a small input change $\mathbf{h}$.

Since $\mathbf{a} = \mathbf{0}$ and $\mathbf{b}_1 = \mathbf{0}$,

$$\mathbf{s_a} = W_1\mathbf{a} + \mathbf{b}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore

$$\mathbf{u_a} = \sigma(\mathbf{s_a}) = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}.$$

Since $\mathbf{b}_2 = \mathbf{0}$,

$$N(\mathbf{a}) = W_2\mathbf{u_a} = \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1/4 \\ 3/4 \end{bmatrix}.$$

Since $\sigma^{'}(0) = 1/4$,

$$D_{\mathbf{a}} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/4 \end{bmatrix}.$$

Therefore

$$
\begin{aligned}
J_N(\mathbf{a}) &= W_2 D_{\mathbf{a}} W_1 \\
&= \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \end{bmatrix}
\begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/4 \end{bmatrix}
\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1/8 & -1/8 \\ 1/8 & 3/8 \end{bmatrix}.
\end{aligned}
$$

$$
N(\mathbf{a}+\mathbf{h}) \approx \begin{bmatrix} 1/4 \\ 3/4 \end{bmatrix} + \begin{bmatrix} 1/8 & -1/8 \\ 1/8 & 3/8 \end{bmatrix} \mathbf{h}.
$$

This gives the intended local prediction: the Jacobian matrix of the composition is a product of local matrices.

Note: Why sigmoid here

The sigmoid function is smooth, so the Jacobian matrix of this block is defined at every input. Some neural-network models use nonsmooth activation rules. Those require extra case analysis. Here the point is the chain rule for Jacobian matrices.

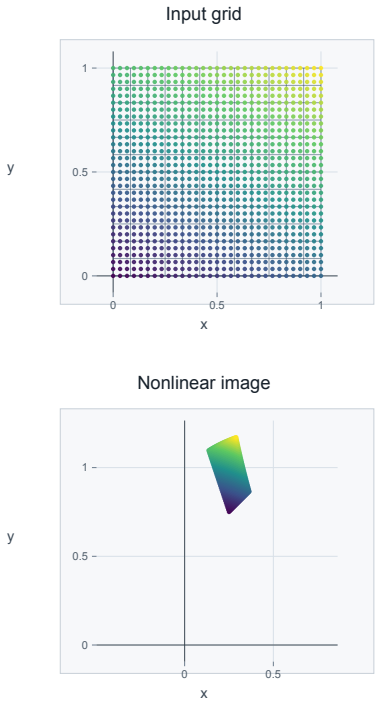


Figure. Square-grid visualization for a two-output sigmoid block.

Activity: A one-query softmax weighting map (U3-LO7, U1-LO8)

A single-query attention-style map can be viewed as a nonlinear vector-valued map.
Let

$$K = \begin{bmatrix} 3 & 0 \\ 0 & 3 \\ -3 & -3 \end{bmatrix}, \qquad V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

For a query vector

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix},$$

define the score vector

$$\mathbf{s} = K\mathbf{q},$$

the softmax weight vector

Activity: A one-query softmax weighting map (U3-LO7, U1-LO8) (continued)

$$\alpha(\mathbf{q}) = \mathrm{softmax}(\mathbf{s}),$$

and the output

$$F(\mathbf{q}) = V^T \alpha(\mathbf{q}).$$

1. Compute $\mathbf{s} = K\mathbf{q}$ in terms of q_1 and q_2.
2. What are the rows of V?
3. Explain why $F(\mathbf{q})$ is a weighted average of the rows of V.
4. Compute $\alpha(\mathbf{0})$ and $F(\mathbf{0})$.
5. Explain why F is not a linear map.
6. What operation makes F nonlinear?

The score vector is

$$\mathbf{s} = K\mathbf{q} = \begin{bmatrix} 3q_1 \\ 3q_2 \\ -3q_1 - 3q_2 \end{bmatrix}.$$

The rows of V are the value vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The softmax vector

$$\alpha(\mathbf{q}) = \begin{bmatrix} \alpha_1(\mathbf{q}) \\ \alpha_2(\mathbf{q}) \\ \alpha_3(\mathbf{q}) \end{bmatrix}$$

has positive entries that add to 1. Therefore

$$F(\mathbf{q}) = V^T \alpha(\mathbf{q}) = \alpha_1(\mathbf{q})\mathbf{v}_1 + \alpha_2(\mathbf{q})\mathbf{v}_2 + \alpha_3(\mathbf{q})\mathbf{v}_3$$

is a weighted average of the value vectors.
At $\mathbf{q} = \mathbf{0}$, the score vector is

$$\mathbf{s} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

All three scores are equal, so

$$\alpha(\mathbf{0}) = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}.$$

Thus

$$F(\mathbf{0}) = V^T \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}.$$

A linear map must send $\mathbf{0}$ to $\mathbf{0}$. Since $F(\mathbf{0}) \neq \mathbf{0}$, the map F is not linear.

The matrix multiplication $K\mathbf{q}$ is linear in $\mathbf{q}$. The final weighted average is linear in the weights if the weights are already fixed. The nonlinear step is the softmax rule that turns scores into weights.

Note: Softmax outputs stay in the value triangle

The output always lies in the triangle spanned by the three value vectors. The query vector does not choose fixed weights. It changes the scores, and the softmax rule changes all weights together.

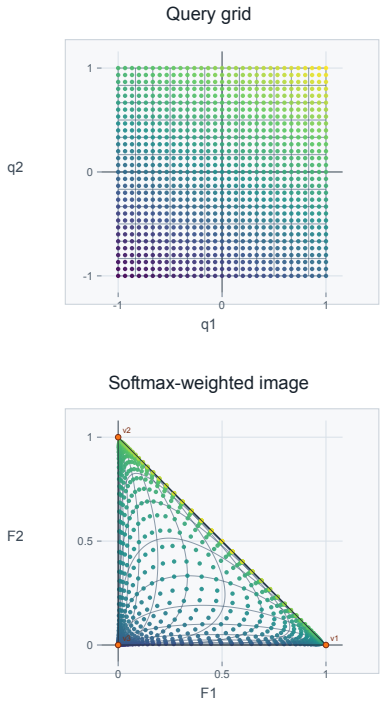


Figure. Square-grid visualization for an attention-style softmax map.

Note: An attention-style nonlinear block

Unit 1 treated attention as three visible operations:

$$\text{dot-product scores} \longrightarrow \text{weights} \longrightarrow \text{weighted averages.}$$

For a sequence of token vectors stored as rows of a matrix X, fixed weight matrices produce

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V.$$

The score matrix is

$$S = QK^T.$$

The entry S_{ij} compares the query for token i with the key for token j. A weighting rule, such as row-wise softmax, converts each row of S into weights. Write the resulting attention-weight matrix as A_{att}. The attention output is

$$H = A_{\text{att}} V.$$

If A_{att} were fixed, then $V \mapsto A_{\text{att}} V$ would be a linear map: each row of H would be a weighted average of rows of V. In attention, A_{att} is not fixed. It is computed from X through Q, K, scores, and softmax. Thus the full rule $X \mapsto H$ is generally nonlinear.

The Unit 3 question is local. Near one fixed input X_0, a nonlinear rule can be approximated by a linear map. After the entries of X and H are arranged as vectors, that local linear map is represented by a Jacobian matrix.

Activity: What depends on the input? (U3-LO5, U3-LO7)

Assume the weight matrices W_Q, W_K, and W_V are fixed. The input is the token matrix X. Define

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V,$$

$$S = QK^T, \qquad A_{\text{att}} = \text{softmax}_{\text{rows}}(S), \qquad H = A_{\text{att}} V.$$

1. Which objects are fixed before the input X is chosen?
2. Which objects are computed from X?
3. Why is $X \mapsto XW_Q$ a linear map when W_Q is fixed?
4. Why does $S = QK^T$ depend on X in two places?
5. If A_{att} were fixed, what would $H = A_{\text{att}} V$ do to the rows of V?
6. In actual attention, why is the full rule $X \mapsto H$ not just one linear map?
7. What Unit 3 object describes the local linear approximation near a fixed input X_0?

The matrices W_Q, W_K, and W_V are fixed. The matrices Q, K, V, S, A_{att}, and H are computed from X.

The rule $X \mapsto XW_Q$ is linear because W_Q is fixed and right multiplication by a fixed matrix distributes over sums and scalar multiples.

The score matrix $S = QK^T$ depends on X through Q and through K. The entries of S are dot products between query rows and key rows, and both sets of rows were computed from X.

If A_{att} were fixed, then $H = A_{\text{att}}V$ would form weighted averages of the rows of V. This is linear in V.

In attention, A_{att} is not fixed. The weights are computed from the input through scores and softmax. The values V also depend on X. Therefore the full rule $X \mapsto H$ is generally nonlinear.

Near a fixed input X_0, the local linear approximation is represented by a Jacobian matrix after the entries of the input and output matrices are

arranged as vectors.

Warning: Specify the input

Before asking whether a rule is linear, first say what is variable and what is fixed. The rule $V \mapsto A_{\text{att}}V$ is linear when A_{att} is fixed. The attention rule $X \mapsto A_{\text{att}}(X)V(X)$ is generally nonlinear because the weights and values both depend on X.

Activity: Comparing square-grid visualizations (U3-LO1, U3-LO5, U3-LO7, U1-LO8)

This unit used three square-grid visualizations: the nonlinear shear from Corners are not enough (U3-LO1), the sigmoid block from A tiny sigmoid block (U3-LO5, U3-LO6, U3-LO7), and the attention-style softmax map from A one-query softmax weighting map (U3-LO7, U1-LO8).
Map A:

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

Map B: a two-output sigmoid block

$$\mathbf{s} = W_1\mathbf{x} + \mathbf{b}_1, \qquad \mathbf{u} = \sigma(\mathbf{s}), \qquad N(\mathbf{x}) = W_2\mathbf{u} + \mathbf{b}_2.$$

Map C: an attention-style softmax map

$$F_{\text{att}}(\mathbf{q}) = V^T \operatorname{softmax}(K\mathbf{q}).$$

1. Which map agrees with a linear shear on the four corners but not at an edge midpoint?

Activity: Comparing square-grid visualizations (U3-LO1, U3-LO5, U3-LO7, U1-LO8) (continued)

2. Which map is built from affine maps with a coordinatewise sigmoid between them?
3. For which map is the local matrix computed by $W_2 D_a W_1$?
4. Which map produces outputs inside the triangle spanned by value vectors?
5. For which map does the nonlinear step come from softmax weights that depend on the input?
6. In all three examples, what does the square-grid visualization show that four corners alone may miss?

Map A is the nonlinear shear. It can agree with a linear shear on selected points but differ between them.
Map B is the sigmoid block. Its Jacobian matrix near a base point has the form $W_2 D_a W_1$, where D_a records the sigmoid derivatives at that base point. The local matrix can change from point to point because the sigmoid derivatives change.
Map C is the attention-style softmax map. Its outputs are weighted averages of the rows of V, so they stay inside the triangle spanned by the

value vectors. The softmax weights depend on the input.
A square-grid visualization can show bending, compression, and changing local behavior that four corners alone may miss.